



F310 Axial Force And Torque Transducer

Geometry: Monocoque strain structure with no moving parts.

The F310 depicted measures force and torque in the same axis with minimal cross talk providing user friendly outputs for direct measurement without mathematical manipulation.

Each channel may be calibrated bi-directionally. The version of the transducer depicted was developed for an OEM producing texture analyser machines (rheometers) for use in the food process industry.

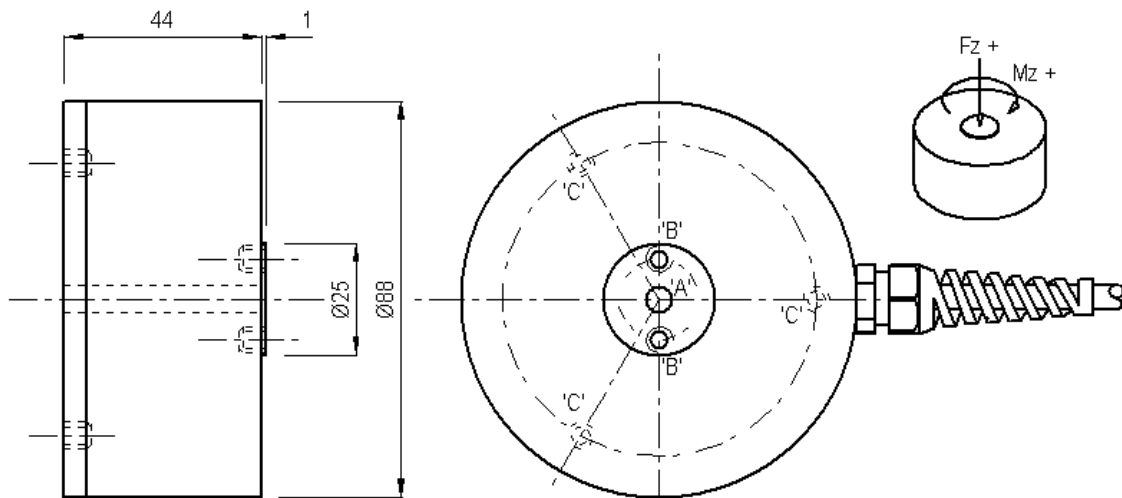
We are happy to design variants of this loadcell to meet your specific requirements. Variants may include different combinations of force and moment axis e.g. $F_x M_z$, $F_y M_x$, $F_z M_y$ etc.

Additional information on specifying a multi-axis loadcell can be found in [Engineering Sheet E015](#).

Features

- ✓ Low cross talk
- ✓ Custom ranges
- ✓ Simple installation
- ✓ Direct outputs without calculation
- ✓ Traceable calibration with certificate included in the standard price
- ✓ Standard 1 year warranty

Outline



'A' Ø6 THRO. 'B' M6x1 ON A 18 PCD - 2 OFF. 'C' M6x1 EQUI SPACED ON A 70 PCD - 3 OFF

Specifications

Parameter	Value	Unit
Non-linearity - Terminal	± 0.05	% RL
Hysteresis	± 0.05	% RL
Creep - 5 minutes	± 0.1	% AL
Repeatability	± 0.05	% RL
Maximum cross talk	1	% RL
Rated output - Nominal	0.6	mV/V
Rated output - Rationalised	0.5	mV/V
Rationalisation tolerance (applies to single direction calibrations)	± 0.5	% RL
Zero load output	± 4	% RL
Temperature effect on rated output per °C	± 0.005	% AL
Temperature effect on zero load output per °C	± 0.02	% RL
Temperature range - Compensated	-10 to +50	°C
Temperature range - Safe	-10 to +80	°C
Excitation voltage - Recommended	10	V
Excitation voltage - Maximum	10	V
Bridge resistance	350	Ω
Insulation resistance - Minimum at 50Vdc	500	M Ω
Overload - Safe	50	% RL
Overload - Ultimate	300	% RL
Weight - Nominal (excluding cable)	250	g

Order Codes

Code	Description
Most F310 loadcells are manufactured to special requirements and are given an F310-Zxxxx number.	

Notes

- AL = Applied load.
- RL = Rated load.
- Temperature coefficients apply over the compensated range.
- Values apply to force and moment in all load combinations.
- Extraneous forces or torques other than those calibrated may create moment errors.

Connections

The F310 is fitted with 2 metres of PVC insulated 9 core screened cable type 7-1-9C. The screen is not connected to the loadcell body.

Function	Wire Colour
	Fz
Excitation +	Red
Excitation -	Blue
Signal +	Yellow
Signal -	Green
Screen	Orange